

# Lovely Professional University, Punjab

| Course Code      | Course Title                                     | Lectures | Tutorials | Practicals | Credits |  |
|------------------|--------------------------------------------------|----------|-----------|------------|---------|--|
| CSE202           | OBJECT ORIENTED PROGRAMMING                      | 3        | 0         | 2          | 4       |  |
| Course Weightage | ATT: 5    CA: 50    ETP: 45                      |          |           |            |         |  |
| Course Focus     | EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP |          |           |            |         |  |

**Course Outcomes :**Through this course students should be able to

CO1 :: identify basic programming constructs and use the newly acquired skills to solve extensive programming problems.

CO2 :: discuss the mechanism of code reusability by creating own libraries of functions.

CO3 :: analyze the logic by designing code capable of passing various test cases.

CO4 :: interpret the principles of the object-oriented model and apply it in the implementation in C++ language.

CO5 :: categorize the theoretical knowledge and insights gained thus far to formulate working code

CO6 :: apply various programming constructs to build software applications

|       | <b>TextBooks ( T )</b>             |               |                |
|-------|------------------------------------|---------------|----------------|
| Sr No | Title                              | Author        | Publisher Name |
| T-1   | OBJECT ORIENTED PROGRAMMING IN C++ | ROBERT LAFORE | PEARSON        |

|       | <b>Reference Books ( R )</b>       |                |                       |
|-------|------------------------------------|----------------|-----------------------|
| Sr No | Title                              | Author         | Publisher Name        |
| R-1   | PROGRAMMING WITH C++               | D RAVICHANDRAN | MCGRAW HILL EDUCATION |
| R-2   | OBJECT ORIENTED PROGRAMMING IN C++ | E BALAGURUSAMY | MCGRAW HILL EDUCATION |

| <b>Relevant Websites ( RW )</b> |                                                                                                   |                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sr No                           | (Web address) (only if relevant to the course)                                                    | Salient Features                                                                                                                                                                  |
| RW-1                            | <a href="http://www.studytonight.com/cpp/">http://www.studytonight.com/cpp/</a>                   | Free website where students can learn the basic concepts of programming and also basics and advanced topics of OOPS. Students can also give the practice tests in this interface. |
| RW-2                            | <a href="https://www.tutorialspoint.com/cplusplus/">https://www.tutorialspoint.com/cplusplus/</a> | Free web site to learn C++ programming for the beginners.                                                                                                                         |
| RW-3                            | <a href="https://www.hackerrank.com/">https://www.hackerrank.com/</a>                             | Programming competitions and contests, programming community                                                                                                                      |
| RW-4                            | <a href="https://codeforces.com/">https://codeforces.com/</a>                                     | Programming competitions and contests, programming community                                                                                                                      |

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| RW-5  | <a href="http://www.codechef.com/">http://www.codechef.com/</a>                                                                                                       | CodeChef hosts Online Programming Competition, Programming Contest                                                                              |
| RW-6  | <a href="https://www.hackerearth.com/">https://www.hackerearth.com/</a>                                                                                               | HackerEarth is a network of top developers across the world. Developers participate in online coding challenges and hackathons, solve problems. |
| RW-7  | <a href="http://www.dailyfreecode.com/Code/perform-array-operations-append-2646.aspx">http://www.dailyfreecode.com/Code/perform-array-operations-append-2646.aspx</a> | Array operations                                                                                                                                |
| RW-8  | <a href="http://www.yolinux.com/TUTORIALS/C++MemoryCorruptionAndMemoryLeaks.html">http://www.yolinux.com/TUTORIALS/C++MemoryCorruptionAndMemoryLeaks.html</a>         | Memory Leak                                                                                                                                     |
| RW-9  | <a href="http://www.learncpp.com/">http://www.learncpp.com/</a>                                                                                                       | learncpp.com is a totally free website devoted to teaching you to program in C++.                                                               |
| RW-10 | <a href="http://cplus.about.com/">http://cplus.about.com/</a>                                                                                                         | About C, C++ and C# brings the latest programming tutorials, programming challenges, C++ for beginners                                          |

| Software/Equipments/Databases |                                                                             |                                                                    |
|-------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|
| Sr No                         | (S/E/D) (only if relevant to the course)                                    | Salient Features                                                   |
| SW-1                          | <a href="http://www.codeblocks.org/home">http://www.codeblocks.org/home</a> | To download Code Blocks Compiler for the execution of C++ programs |
| SW-2                          | <a href="https://www.bloodshed.net/">https://www.bloodshed.net/</a>         | To download DEV- Cpp for the execution of C++ programs             |

| Virtual Labs ( VL ) |                                                                                         |                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Sr No               | (VL) (only if relevant to the course)                                                   | Salient Features                                                                            |
| VL-1                | <a href="http://sourceforge.net/projects/vle/">http://sourceforge.net/projects/vle/</a> | Students need to download the system and study the slightly complex programming environment |

| LTP week distribution: (LTP Weeks) |   |
|------------------------------------|---|
| Weeks before MTE                   | 7 |
| Weeks After MTE                    | 7 |
| Spill Over (Lecture)               | 7 |

## Detailed Plan For Lectures

| Week Number | Lecture Number | Broad Topic(Sub Topic)                                                                                           | Chapters/Sections of Text/reference books | Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs | Lecture Description                                                                   | Learning Outcomes                                                                           | Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned | Live Examples |
|-------------|----------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------|
| Week 1      | Lecture 1      | Concepts and Basics of C++ Programming(Differences between procedural and object oriented programming paradigms) | T-1                                       | RW-9<br>SW-2                                                                    | Prerequisites of C plus Plus.<br>Delivery of lecture 0 and introduction to the course | Students will come to Know the difference between procedure and object oriented programming | Discussion, Real life problems                                                     |               |

|        |           |                                                                                                          |            |               |                                                                                                                                 |                                                                                                            |                                                                                   |                                                                                                    |
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| Week 1 | Lecture 1 | Concepts and Basics of C++ Programming(Features of Input/output Streams)                                 | T-1        | RW-9<br>SW-2  | Prerequisites of C plus Plus.<br>Delivery of lecture 0 and introduction to the course                                           | Students will come to Know the difference between procedure and object oriented programming                | Discussion, Real life problems                                                    |                                                                                                    |
|        | Lecture 2 | Concepts and Basics of C++ Programming(Reading and writing data using cin and cout)                      | T-1        | RW-9          | Practice of programs using cin and cout                                                                                         | Students will come to know about cin and cout statements                                                   | Practice Problem-Solving(Third party platform)                                    |                                                                                                    |
|        | Lecture 3 | Concepts and Basics of C++ Programming(Creating classes, Class objects, Accessing class members)         | T-1        | RW-9          | creating classes, class objects and accessing class members                                                                     | The students will also know how to create classes and its objects                                          | Practice Problem-Solving(Third party platform)                                    | An example of car can be taken which is a class and a particular model of the car can be an object |
| Week 2 | Lecture 4 | Concepts and Basics of C++ Programming(Differences between Structures, Unions, Enumerations and Classes) | T-1<br>R-2 | RW-9<br>RW-10 | Description of structures, unions and classes, Enumeration                                                                      | Students will learn as how to group multiple data as a single entity and the differences between them      | Discussion, Real life problems and Practice Problem-Solving(Third party platform) | A nursery having different variety of flowers can be considered as a class, structure              |
|        | Lecture 5 | Concepts and Basics of C++ Programming(Inline and Non-inline member functions)                           | T-1<br>R-2 |               | To Understand and use concept of inline and non-inline member functions,static data members and static member Functions         | Students will learn the use of inline functions,static data members and static member functions            | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |                                                                                                    |
|        |           | Concepts and Basics of C++ Programming(Static data members and static member functions)                  | T-1<br>R-2 |               | To Understand and use concept of inline and non-inline member functions,static data members and static member Functions         | Students will learn the use of inline functions,static data members and static member functions            | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |                                                                                                    |
|        | Lecture 6 | Functions(Functions with Default parameters/arguments)                                                   | T-1        | RW-1<br>RW-9  | How to use Functions with default parameters,Displaying output in various forms and using inline functions for faster execution | Students will understand as how to use functions with default parameters and understand cascading concepts | Use Learning resource (Third party platform)                                      |                                                                                                    |

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| Week 2 | Lecture 6 | Functions(Inline Functions)                                     | T-1 | RW-1<br>RW-9 | How to use Functions with default parameters,Displaying output in various forms and using inline functions for faster execution | Students will understand as how to use functions with default parameters and understand cascading concepts | Use Learning resource (Third party platform) |                                                                                                                             |
|        |           | Functions(Manipulator Functions)                                | T-1 | RW-1<br>RW-9 | How to use Functions with default parameters,Displaying output in various forms and using inline functions for faster execution | Students will understand as how to use functions with default parameters and understand cascading concepts | Use Learning resource (Third party platform) |                                                                                                                             |
| Week 3 | Lecture 7 | Functions(Function overloading and Scope rules)                 | T-1 | RW-9         | Details about function overloading, local and global scope variables and friend function , friend function and friend class     | Students will understand the advantages of using function overloading, friend function and friend class    | Use Learning resource (Third party platform) | A friend has access to your games and some of your possessions which can be used for friend function                        |
|        |           | Functions(Friend of a class (friend function and friend class)) | T-1 | RW-9         | Details about function overloading, local and global scope variables and friend function , friend function and friend class     | Students will understand the advantages of using function overloading, friend function and friend class    | Use Learning resource (Third party platform) | A friend has access to your games and some of your possessions which can be used for friend function                        |
|        | Lecture 8 | Functions(Reference variables)                                  | T-1 | RW-9         | Details about the differences in various function calls and recursion                                                           | Students will understand the differences between various function calls, and recursion                     | Use Learning resource (Third party platform) | A record playing from the beginning to end can be used as an example of Recursion as function and member function of class. |

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| Week 3 | Lecture 8 | Functions(Differences between Call by value)                                                                    | T-1 | RW-9         | Details about the differences in various function calls and recursion                                                              | Students will understand the differences between various function calls, and recursion                | Use Learning resource (Third party platform) | A record playing from the beginning to end can be used as an example of Recursion as function and member function of class. |
|        |           | Functions(Call by address and call by reference)                                                                | T-1 | RW-9         | Details about the differences in various function calls and recursion                                                              | Students will understand the differences between various function calls, and recursion                | Use Learning resource (Third party platform) | A record playing from the beginning to end can be used as an example of Recursion as function and member function of class. |
|        |           | Functions(Recursion (Function, Member Function))                                                                | T-1 | RW-9         | Details about the differences in various function calls and recursion                                                              | Students will understand the differences between various function calls, and recursion                | Use Learning resource (Third party platform) | A record playing from the beginning to end can be used as an example of Recursion as function and member function of class. |
|        | Lecture 9 | Pointers, Reference Variables, Arrays and String Concepts(Void pointer, Pointer arithmetic, Pointer to pointer) | T-1 | RW-2<br>RW-9 | Usage of reference variables and differences from pointers, void pointer and arithmetic operations on pointers, pointer to pointer | The students will learn about the various arithmetic operations on pointers and the types of pointers | Use Learning resource (Third party platform) | A street sign pointing to the location of a bus terminus could be an example of a pointer                                   |
|        |           | Pointers, Reference Variables, Arrays and String Concepts(Differences between pointer and reference variables)  | T-1 | RW-2<br>RW-9 | Usage of reference variables and differences from pointers, void pointer and arithmetic operations on pointers, pointer to pointer | The students will learn about the various arithmetic operations on pointers and the types of pointers | Use Learning resource (Third party platform) | A street sign pointing to the location of a bus terminus could be an example of a pointer                                   |

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| Week 4 | Lecture 10 | Pointers, Reference<br><br>Variables, Arrays and String Concepts(Possible problems with the use of pointers - Dangling pointer, Wild pointer, Null pointer assignment) | T-1<br><br>R-1<br>R-2 | RW-1<br><br>RW-8                                 | Various pointers and types of problems which may arise due to the use of pointers classes having pointers as members and this pointer | The students will learn about the various pointers and problems associated with pointer use and use of pointer with object of class | Use Learning resource (Third party platform)                              | A street sign pointing to the location of a bus terminus could be an example of pointer            |
|        |            | Pointers, Reference<br>Variables, Arrays and String Concepts(Classes containing pointers, Pointer to objects, this pointer)                                            | T-1<br>R-1<br>R-2     | RW-1<br>RW-8                                     | Various pointers and types of problems which may arise due to the use of pointers classes having pointers as members and this pointer | The students will learn about the various pointers and problems associated with pointer use and use of pointer with object of class | Use Learning resource (Third party platform)                              | A street sign pointing to the location of a bus terminus could be an example of pointer            |
|        | Lecture 11 | Pointers, Reference<br>Variables, Arrays and String Concepts(Array declaration and processing of multidimensional arrays (inside main and inside class))               | T-1                   | RW-4<br>RW-5<br>RW-6<br>RW-7<br><br>RW-9<br>SW-1 | Declaration and processing of multidimensional arrays and use of array of objects                                                     | Students will learn to use multidimensional arrays along with the creation of 1D and multidimensional array of objects              | Practice Problem-Solving and Use Learning resource (Third party platform) | Rows of students all belonging to the same section can be used to explain a multidimensional array |
|        |            | Pointers, Reference<br>Variables, Arrays and String Concepts(Pointer to data member)                                                                                   | T-1                   | RW-4<br>RW-5<br>RW-6<br>RW-7<br><br>RW-9<br>SW-1 | Declaration and processing of multidimensional arrays and use of array of objects                                                     | Students will learn to use multidimensional arrays along with the creation of 1D and multidimensional                               | Practice Problem-Solving and Use Learning resource (Third party platform) | Rows of Students all belonging to the same section can be used to explain a Multidimensional L     |
|        |            |                                                                                                                                                                        |                       |                                                  |                                                                                                                                       |                                                                                                                                     |                                                                           |                                                                                                    |

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|        |            |                                                                                                                                                  |     |      |                                                                                                        | array of objects                                                                                                         |                                                                           | Array                                                                                                 |
|        | Lecture 12 | Pointers, Reference Variables, Arrays and String Concepts(Array of objects, The Standard C++ string class-defining and assigning string objects) | T-1 | RW-9 | Using C++ string class, operations on strings                                                          | Students will learn to use the string class in C++ effectively                                                           | Practice Problem-Solving and Use Learning resource (Third party platform) | The address of anyone's home contains a mix of characters, digits etc and is an example of String     |
| Week 5 | Lecture 13 |                                                                                                                                                  |     |      | Test - Code based 1                                                                                    |                                                                                                                          |                                                                           |                                                                                                       |
|        | Lecture 14 | Constructors, Destructors and File Handling(Manager functions (constructors and destructor), Default constructor)                                | T-1 | RW-9 | Manager functions involving various types of constructors, their use and differences                   | Students will learn about the difference between various types of constructors and their use                             | Use Learning resource (Third party platform)                              | The way of assigning a value to a variable of built in data type can be used to describe constructors |
|        | Lecture 15 | Constructors, Destructors and File Handling(Manager functions (constructors and destructor), Default constructor)                                | T-1 | RW-9 | Manager functions involving various types of constructors, their use and differences                   | Students will learn about the difference between various types of constructors and their use                             | Use Learning resource (Third party platform)                              | The way of assigning a value to a variable of built in data type can be used to describe constructors |
| Week 6 | Lecture 16 | Constructors, Destructors and File Handling (Constructor with default arguments, Destructors)                                                    | T-1 | RW-9 | Manager functions involving various types of constructors, their use and differences.use of destructor | Students will learn about the difference between various types of constructors and the use of constructor and destructor | Use Learning resource (Third party platform)                              | The way of assigning a value to a variable of built in data type can be used to describe constructors |
|        |            | Constructors, Destructors and File Handling (Parameterized constructor, Copy constructor , Initializer lists)                                    | T-1 | RW-9 | Manager functions involving various types of constructors, their use and differences.use of destructor | Students will learn about the difference between various types of constructors and the use of constructor and destructor | Practice Problem-Solving and Use Learning resource (Third party platform) | The way of assigning a value to a variable of built in data type can be used to describe constructors |

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|        | Lecture 17 | Data File operations<br>(Opening and closing of files, Modes of file)                              | T-1               | RW-10 | Use of file handling in C ++, opening and closing files, modes of opening files, reading and writing of files | Students will learn how to open the files, the various modes to open and how to read and write into files                      | Practice Problem-Solving and Use Learning resource (Third party platform) | The file folder which is used to hold the various documents of a person can be used to explain the concept of file handling |
| Week 6 | Lecture 17 | Data File operations(File stream functions, Reading/Writing of files)                              | T-1               | RW-10 | Use of file handling in C ++, opening and closing files, modes of opening files, reading and writing of files | Students will learn how to open the files, the various modes to open and how to read and write into files                      | Practice Problem-Solving and Use Learning resource (Third party platform) | The file folder which is used to hold the various documents of a person can be used to explain the concept of file handling |
|        | Lecture 18 | Data File operations (Sequential access and random access file processing, Binary file operations) | T-1<br>R-1<br>R-2 | RW-9  | Differences between sequential and random access                                                              | Students will learn the difference between sequential and random access and to use file handling in an object oriented program | Practice Problem-Solving and Use Learning resource (Third party platform) | The file folder which is used to hold the various documents of a person can be used to explain the concept of file handling |
|        |            | Data File operations(Classes and file operations, Structures and file operation)                   | T-1<br>R-1<br>R-2 | RW-9  | Differences between sequential and random access                                                              | Students will learn the difference between sequential and random access and to use file handling in an object oriented program | Practice Problem-Solving and Use Learning resource (Third party platform) | The file folder which is used to hold the various documents of a person can be used to explain the concept of file handling |
| Week 7 | Lecture 19 | Data File operations(Classes and file operations, Structures and file operation)                   | T-1<br>R-1<br>R-2 | RW-9  | Differences between sequential and random access                                                              | Students will learn the difference between sequential and random access and to use file handling in an object oriented program | Practice Problem-Solving and Use Learning resource (Third party platform) | The file folder which is used to hold the various documents of a person can be used to explain the concept of file handling |
|        |            | <b>SPILL OVER</b>                                                                                  |                   |       |                                                                                                               |                                                                                                                                |                                                                           |                                                                                                                             |
| Week 7 | Lecture 20 |                                                                                                    |                   |       | Spill Over                                                                                                    |                                                                                                                                |                                                                           |                                                                                                                             |
|        | Lecture 21 |                                                                                                    |                   |       | Spill Over                                                                                                    |                                                                                                                                |                                                                           |                                                                                                                             |
|        |            | <b>MID-TERM</b>                                                                                    |                   |       |                                                                                                               |                                                                                                                                |                                                                           |                                                                                                                             |



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| Week 8  | Lecture 22 | Operator Overloading and Type Conversion(Operator Overloading (unary operator, binary operator overloading))    | T-1               | RW-1<br>RW-2<br>RW-9 | Usage and program of overloading unary operators in object oriented programming                       | Students will learn to overload unary operators in an object oriented program                             | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
| Week 8  | Lecture 23 | Operator Overloading and Type Conversion(Operator Overloading (unary operator, binary operator overloading))    | T-1               | RW-1<br>RW-2<br>RW-9 | Usage and program of overloading unary operators in object oriented programming                       | Students will learn to overload unary operators in an object oriented program                             | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
|         | Lecture 24 | Operator Overloading and Type Conversion(Type conversions - basic type to class type, class type to basic type) | T-1<br>R-1<br>R-2 | RW-9                 | Conversion of a user defined data type to inbuilt data type and vice versa                            | Students will learn to convert user defined data type into built in type and inbuilt to user defined type | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
| Week 9  | Lecture 25 | Operator Overloading and Type Conversion(Type conversions - basic type to class type, class type to basic type) | T-1<br>R-1<br>R-2 | RW-9                 | Conversion of a user defined data type to inbuilt data type and vice versa                            | Students will learn to convert user defined data type into built in type and inbuilt to user defined type | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
|         | Lecture 26 | Inheritance(Inheritance Basics – derived class and base class)                                                  | T-1<br>R-1<br>R-2 | RW-2                 | Inheritance importance and concept of base and derived class and different modes to perform the same. | Students will learn about the importance of inheritance and its modes                                     | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
|         |            | Inheritance(Modes (private, protected, public inheritance))                                                     | T-1<br>R-1<br>R-2 | RW-2                 | Inheritance importance and concept of base and derived class and different modes to perform the same. | Students will learn about the importance of inheritance and its modes                                     | Practice Problem-Solving and Use Learning resource (Third party platform) |  |
|         | Lecture 27 | Inheritance(Types (simple, multi-level, multiple and hierarchical))                                             | T-1<br>R-1<br>R-2 | RW-2                 | various Inheritance types (simple,multi-level, multiple and hierarchical)                             | Students will learn about the importance of inheritance and its types                                     | Use Learning resource (Third party platform)                              |  |
| Week 10 | Lecture 28 | Inheritance(Overriding member functions)                                                                        | T-1<br>R-1<br>R-2 | RW-2                 | How to override member functions                                                                      | Students will learn about overriding of member functions                                                  | Use Learning resource (Third party platform)                              |  |

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| Week 10 | Lecture 29 | Inheritance(Order of execution of constructors and destructors)                                      | T-1               | RW-2 | The execution order of constructors and destructors, Various ambiguities that may arise in inheritance and the virtual base class. The execution order of constructors and destructors, Various ambiguities that may arise in inheritance and the virtual base class | Students will learn about the order of execution of constructors and destructors, how to resolve various ambiguities in inheritance and the importance of virtual base class | Use Learning resource (Third party platform)                                      |  |
|         |            | Inheritance(Resolving ambiguities in inheritance, Virtual base class)                                | T-1               | RW-2 | The execution order of constructors and destructors, Various ambiguities that may arise in inheritance and the virtual base class. The execution order of constructors and destructors, Various ambiguities that may arise in inheritance and the virtual base class | Students will learn about the order of execution of constructors and destructors, how to resolve various ambiguities in inheritance and the importance of virtual base class | Use Learning resource (Third party platform)                                      |  |
|         | Lecture 30 | Inheritance(Aggregation and Composition.)                                                            | T-1<br>R-1<br>R-2 | RW-2 | types of association which are used to represent the relationship between two classes                                                                                                                                                                                | Students will learn about the both Composition and Aggregation are types of association which are used to represent the relationship between two classes.                    | Use Learning resource (Third party platform)                                      |  |
| Week 11 | Lecture 31 |                                                                                                      |                   |      | Test - Code based 2                                                                                                                                                                                                                                                  |                                                                                                                                                                              |                                                                                   |  |
|         | Lecture 32 | Dynamic Memory Management and Polymorphism(Dynamic memory allocation using new and delete operators) | T-1<br>R-1        | RW-9 | Allocation and deallocation of memory                                                                                                                                                                                                                                | Students will learn about allocating and deallocating memory                                                                                                                 | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |

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| Week 11 | Lecture 32 | Dynamic Memory Management and Polymorphism(Memory leak and allocation failures)                                       | T-1<br>R-1        | RW-9         | Allocation and deallocation of memory                                                                                          | Students will learn about allocating and deallocating memory                                                                                                               | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|         | Lecture 33 | Dynamic Memory Management and Polymorphism(Virtual destructors, Compile and run time polymorphism, Virtual functions) | T-1               | RW-9<br>SW-1 | importance of virtual Destructors, An OOPS concept, difference between early and late binding, importance of virtual Functions | Students will learn the importance of virtual destructors, Students will learn the distinction between compile time and run time polymorphism and use of virtual functions | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|         |            | Dynamic Memory Management and Polymorphism(Pure virtual functions)                                                    | T-1               | RW-9<br>SW-1 | importance of virtual Destructors, An OOPS concept, difference between early and late binding, importance of virtual Functions | Students will learn the importance of virtual destructors, Students will learn the distinction between compile time and run time polymorphism and use of virtual functions | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
| Week 12 | Lecture 34 | Dynamic Memory Management and Polymorphism(Abstract classes and concrete class)                                       | T-1<br>R-1<br>R-2 | RW-9<br>VL-1 | Difference between abstract and concrete class, Use of self-referential class                                                  | Students will learn The abstract and concrete class, self-referential class                                                                                                | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|         |            | Dynamic Memory Management and Polymorphism(Introduction to Self-Referential class)                                    | T-1<br>R-1<br>R-2 | RW-9<br>VL-1 | Difference between abstract and concrete class, Use of self-referential class                                                  | Students will learn The abstract and concrete class, self-referential class                                                                                                | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|         | Lecture 35 | Dynamic Memory Management and Polymorphism(Dynamic constructors)                                                      | T-1<br>R-1<br>R-2 | RW-1<br>RW-9 | difference between early and late binding, Use of dynamic constructors                                                         | Students will learn the difference between early and late binding, the use of dynamic constructors                                                                         | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |

|                   |            |                                                                                                                                                                                             |                   |                      |                                                                                                          |                                                                                                               |                                                                                   |  |
|-------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--|
| Week 12           | Lecture 35 | Dynamic Memory Management and Polymorphism(Early binding and late binding)                                                                                                                  | T-1<br>R-1<br>R-2 | RW-1<br>RW-9         | difference between early and late binding, Use of dynamic constructors                                   | Students will learn the difference between early and late binding, the use of dynamic constructors            | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|                   | Lecture 36 |                                                                                                                                                                                             |                   |                      | Test - Code based 3                                                                                      |                                                                                                               |                                                                                   |  |
| Week 13           | Lecture 37 | Exception Handling, Templates and Standard Template Library (STL) (Basics of exception handling, Exception handling mechanism)                                                              | T-1<br>R-1<br>R-2 | RW-2<br>RW-3<br>RW-9 | Usage of the important concept of exception handling and its importance in handling erroneous conditions | Students will learn to deal with error conditions and handle them                                             | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|                   |            | Exception Handling, Templates and Standard Template Library (STL) (Throwing mechanism, Catching mechanism)                                                                                  | T-1<br>R-1<br>R-2 | RW-2<br>RW-3<br>RW-9 | Usage of the important concept of exception handling and its importance in handling erroneous conditions | Students will learn to deal with error conditions and handle them                                             | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
|                   | Lecture 38 |                                                                                                                                                                                             |                   |                      | Programming Practice                                                                                     |                                                                                                               |                                                                                   |  |
|                   | Lecture 39 | Exception Handling, Templates and Standard Template Library (STL) (Rethrowing an exception, Function template and class template)                                                           | T-1<br>R-1<br>R-2 | RW-2<br>RW-9         | Use of function and class template                                                                       | student will learn how to deal with function and class template                                               | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
| Week 14           | Lecture 40 | Exception Handling, Templates and Standard Template Library (STL) (Class template with inheritance, Introduction to STL- Containers, Algorithms and iterators, Container - Vector and List) | T-1<br>R-2        | RW-1<br>RW-9         | Introduction to STL□Containers, Algorithms and Iterators, Container–Vector and List                      | Students will learn about the concept STL- Containers, Algorithms and iterators, Container - Vector and List) | Discussion, Real life problems and Practice Problem-Solving(Third party platform) |  |
| <b>SPILL OVER</b> |            |                                                                                                                                                                                             |                   |                      |                                                                                                          |                                                                                                               |                                                                                   |  |
| Week 14           | Lecture 41 |                                                                                                                                                                                             |                   |                      | Spill Over                                                                                               |                                                                                                               |                                                                                   |  |
|                   | Lecture 42 |                                                                                                                                                                                             |                   |                      | Spill Over                                                                                               |                                                                                                               |                                                                                   |  |
| Week 15           | Lecture 43 |                                                                                                                                                                                             |                   |                      | Spill Over                                                                                               |                                                                                                               |                                                                                   |  |
|                   | Lecture 44 |                                                                                                                                                                                             |                   |                      | Spill Over                                                                                               |                                                                                                               |                                                                                   |  |
|                   | Lecture 45 |                                                                                                                                                                                             |                   |                      | Spill Over                                                                                               |                                                                                                               |                                                                                   |  |

## Scheme for CA:

CA Category of this Course Code is:C020102 (Total 4 tasks, 2 compulsory and out of remaining 1 best out of 2 to be considered)

| Component            | Iscompulsory | Weightage (%) | Mapped CO(s)                 |
|----------------------|--------------|---------------|------------------------------|
| Test - Code based 1  | NO           | 30            | CO1, CO2, CO3                |
| Test - Code based 2  | NO           | 30            | CO3, CO4, CO5, CO6           |
| Test - Code based 3  | Yes          | 40            | CO3, CO4, CO5, CO6           |
| Programming Practice | Yes          | 30            | CO1, CO2, CO3, CO4, CO5, CO6 |

## Details of Academic Task(s)

| Academic Task        | Objective                                 | Detail of Academic Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Nature of Academic Task (group/individuals) | Academic Task Mode | Marks | Allotment / submission Week |
|----------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------|-------|-----------------------------|
| Programming Practice | To evaluate student's overall performance | The students will be given approx. 132 coding problems and 90 MCQs on the selected third party platform. The questions will be framed with equal distribution from the complete syllabus. In order to qualify for programming practice marks, the student should solve at least 50% of the coding problems and 50% of MCQ questions (eligibility condition). The maximum marks out of 15 marks for which the student would be eligible for Programming Practice would be based on the Percentage of questions solved by the student<br>Example- If a student solves 106 questions out of 132 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). The final marks for Programming Practice would be calculated by prorating the eligible marks for which the student is eligible (as explained in the above point) with the percentage of marks the student has scored in the proctored Coding Contests conducted as CA's along with the mandatory written test. (The final marks would be rounded up for the students). Example – If a student solves 106 questions out of 132 questions (i.e. 80% of questions solved) then the student would be eligible for 80% of 15 marks which is 12 marks (round-up would be used in case of decimal values). And if the student has scored 24 out of 30 in the other CA's i.e. 80% marks in CA, his Programming practice final marks would be 80% of 12 marks that he was eligible for which is 9.6 rounded up to 10 marks out of 15 for Programming Practice. | Individual                                  | Online             | 15    | 2 / 13                      |
| Test - Code based 1  | To ensure understanding of the            | Mix of MCQs (10 marks) and Coding Problems(20 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Individual                                  | Online             | 30    | 5 / 6                       |

|                     |                                                                                                                  |                                                                              |            |         |    |         |
|---------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------|---------|----|---------|
|                     | concepts and check the student's progress and his performance on individual basis                                |                                                                              |            |         |    |         |
| Test - Code based 2 | To ensure understanding of the concepts and check the student's progress and his performance on individual basis | Mix of MCQs (10 marks) and Coding Problems(20 marks)                         | Individual | Online  | 30 | 8 / 10  |
| Test - Code based 3 | To ensure understanding of the concepts and check the student's progress and his performance on individual basis | The test will be of one hour duration and will be conducted in offline mode. | Individual | offline | 30 | 12 / 13 |

### Detailed Plan For Practicals

| Practical No | Broad topic                       | Subtopic                                                                                           | Other Readings | Learning Outcomes                                                                 |
|--------------|-----------------------------------|----------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------|
| Practical 1  | List of Practicals / Experiments: | Concepts and Basics of C++ Programming:<br>Programs to define classes and structures               |                | Students will able to understand the concept of class and object                  |
| Practical 2  | List of Practicals / Experiments: | Concepts and Basics of C++ Programming:<br>Programs to define classes and structures               |                | Students will able to understand the concept of class and object                  |
| Practical 3  | List of Practicals / Experiments: | Program to demonstrate inline, non inline member functions and Static function                     |                | Students will able to understand the concept of class and object, static function |
| Practical 4  | List of Practicals / Experiments: | Functions :<br>Program to implement function overloading, friend function and friend class         |                | Students will able to understand the concept of class and object, static function |
| Practical 5  | List of Practicals / Experiments: | Program to demonstrate the difference between call by value, call by address and call by reference |                | Students will able to understand the concept of class and object                  |

|              |                                   |                                                                                                                         |  |                                                                                          |
|--------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------|
| Practical 6  | List of Practicals / Experiments: | Pointers, Reference Variables, Arrays and String Concepts:<br>Program to demonstrate the type of pointers               |  | Students will be able to understand the concept of pointers                              |
| Practical 7  | List of Practicals / Experiments: | Program to process multidimensional array and array of objects                                                          |  | Students will be able to understand the concept of array and array of objects            |
| Practical 8  | List of Practicals / Experiments: | Constructors, Destructors and File Handling:<br>Program to demonstrate constructor, destructor and type of constructors |  | Students will able to understand the concept of constructor and destructor               |
| Practical 9  | List of Practicals / Experiments: | Constructors, Destructors and File Handling:<br>Program to demonstrate constructor, destructor and type of constructors |  | Students will able to understand the concept of constructor and destructor               |
| Practical 10 | List of Practicals / Experiments: | Data File operations:<br>Program to demonstrate the modes of file, Program to demonstrate type of files                 |  | Students will able to understand the concept of files<br><br>in c++                      |
| Practical 11 | List of Practicals / Experiments: | Data File operations:<br>Program to demonstrate the modes of file, Program to demonstrate type of files                 |  | Students will able to understand the concept of files<br><br>in c++                      |
| Practical 12 | List of Practicals / Experiments: | Operator Overloading and Type Conversion:<br>Program to demonstrate the operator overloading and type conversion        |  | Students will able to understand the concept of operator overloading and type conversion |
| Practical 13 | List of Practicals / Experiments: | Operator Overloading and Type Conversion:<br>Program to demonstrate the operator overloading and type conversion        |  | Students will able to understand the concept of operator overloading and type conversion |
| Practical 14 | List of Practicals / Experiments: | Operator Overloading and Type Conversion:<br>Program to demonstrate the operator overloading and type conversion        |  | Students will able to understand the concept of operator overloading and type conversion |
| Practical 15 | List of Practicals / Experiments: | Inheritance:<br>Program to demonstrate the type of inheritance                                                          |  | Students will able to understand the concept of inheritance and the types of inheritance |
| Practical 16 | List of Practicals / Experiments: | Inheritance:<br>Program to demonstrate the type of inheritance                                                          |  | Students will able to understand the concept of inheritance and the types of inheritance |
| Practical 17 | List of Practicals / Experiments: | Program to demonstrate the ambiguities in inheritance                                                                   |  | Students will able to understand the concept of inheritance and the types of inheritance |

|              |                                   |                                                                                                               |  |                                                                                                                            |
|--------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------|
| Practical 18 | List of Practicals / Experiments: | Program to demonstrate the ambiguities in inheritance                                                         |  | Students will able to understand the concept of inheritance and the types of inheritance                                   |
| Practical 19 | List of Practicals / Experiments: | Dynamic Memory Management and Polymorphism : Program to use new and delete for dynamic memory management      |  | Students will able to understand the concept of Dynamic memory allocation and its advantages over static memory allocation |
| Practical 20 | List of Practicals / Experiments: | Dynamic Memory Management and Polymorphism : Program to use new and delete for dynamic memory management      |  | Students will able to understand the concept of Dynamic memory allocation and its advantages over static memory allocation |
| Practical 21 | List of Practicals / Experiments: | Program to demonstrate the compile time and run time polymorphism                                             |  | Students will able to understand the concept of Dynamic memory allocation and its advantages over static memory allocation |
| Practical 22 | List of Practicals / Experiments: | Program to demonstrate abstract class and dynamic constructor                                                 |  | Students will able to understand the concept of Dynamic memory allocation and its advantages over static memory allocation |
| Practical 23 | List of Practicals / Experiments: | Exception Handling, Templates and Standard Template Library (STL) : Program to demonstrate exception handling |  | Students will able to understand the concept of exception handling and templates in c++                                    |
| Practical 24 | List of Practicals / Experiments: | Exception Handling, Templates and Standard Template Library (STL) : Program to demonstrate exception handling |  | Students will able to understand the concept of exception handling and templates in c++                                    |
| Practical 25 | List of Practicals / Experiments: | Program to demonstrate function template and class template                                                   |  | Students will able to understand the concept of exception handling and templates in c++                                    |
| Practical 26 | List of Practicals / Experiments: | Program to demonstrate function template and class template                                                   |  | Students will able to understand the concept of exception handling and templates in c++                                    |
| Practical 27 | List of Practicals / Experiments: | Program to demonstrate STL-Containers, Algorithms and Iterators.                                              |  | Students will able to understand the concept of STL.                                                                       |
| Practical 28 | List of Practicals / Experiments: | Program to demonstrate STL-Containers, Algorithms and Iterators.                                              |  | Students will able to understand the concept of STL                                                                        |
|              | <b>SPILL OVER</b>                 |                                                                                                               |  |                                                                                                                            |
| Practical 29 | Spill Over                        |                                                                                                               |  |                                                                                                                            |